

# DEVA ANAND

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## Education

**MS, Computer Science**, University of Massachusetts Amherst 2025–2027

*Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering*

**BE, Computer Science and Engineering**, Vels Institute of Science, Technology and Advanced Studies 2019–2023

## Professional Experience

**Wysa**, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients**, processing **10,000+ monthly submissions** via REST integrations; contributed to **\$340K+** in revenue.
- Architected a full-stack platform for creating and evaluating **ML training data** (React, Node.js, MongoDB) used daily by the AI team, cutting manual annotation by **100+ hours/month** and lifting labeling throughput by **60%**.
- Implemented backend reliability and security controls, AWS KMS encryption, auth hardening, rate limiting, and integration tests, remediating **20+ VAPT findings**.
- Piloted a COO-sponsored GitHub Copilot experiment, evaluating AI-generated code across tech-debt PRs, code migrations, and fixes as primary technical POC, and onboarding a second developer.

**Cario Growth Services**, Backend Developer Intern | *Chennai, India* Nov 2022 – May 2023

- Shipped a production LLM feature (auto-commenting via **4-bit quantized Llama-7B, llama.cpp**) with per-persona **prompt workflows**, async queued generation, and a content-safety filter, built end to end with no ML mentor.
- Designed and shipped **20+ RESTful APIs** (Node.js, Fastify, PostgreSQL) with validation and clear contracts used in production, and migrated core repositories from JavaScript to TypeScript.

**Freelance / Open Source**, Full-Stack Engineer | *Remote* Jul 2022 – Aug 2023

- Built and deployed **Underdogs Fitness** (Website), a production MERN platform with Stripe payments, cron automation, and multi-branch support, independently end to end; live **3+ years**.

## Projects

**AgenticSearch – Provenance-First Entity Discovery** (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a multi-stage agent that breaks open-ended queries into steps via typed retrieval planning, tool use, and deterministic extraction with LLM fallback and **cell-level provenance**; hardened with **150+ tests** and provider routing across OpenAI/Groq.

**AutoEval – Retrieval Evaluation Improvement Agent** (GitHub) Python, IR evaluation, Git automation

- Built an evaluation-improvement agent that proposes queries and labels, scores them against multiple systems, and accepts only updates that raise discriminative power, with anti-Goodharting guardrails (schema checks, duplicate rejection, read-only holdout protection).

**cf-ai-research-scout – Edge-Native AI Assistant** (GitHub) TypeScript, Cloudflare Workers AI, Durable Objects, D1

- Built an agent with durable **memory** and multi-step **orchestration** (Durable Objects, WorkflowEntrypoint) that uses tools to fetch sources and generate research digests, surviving process restarts and individual tool failures.

## Publications

- “Alzheimer’s Disease Classification using Transfer Learning,” IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

## Skills

**Languages & Databases:** Python, TypeScript, JavaScript, SQL, MongoDB, PostgreSQL, SQLite

**LLM & Agents:** LLM agents, tool use, task planning, orchestration, memory, prompt engineering, evaluation

**ML & NLP:** PyTorch, Hugging Face Transformers, NLP, deep learning, classification, model evaluation

**Web & Deployment:** React, Next.js, Node.js, FastAPI, Docker, REST APIs, Google Cloud Run

**Cloud & Platforms:** AWS, GCP, Azure, Linux, MongoDB Atlas

## Achievements

- Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.